5to. Congreso Internacional de Tecnología y Ciencias Aplicadas (CITCA)

T e c N M / C E N I D E T

Fecha: 27 de noviembre 2025

Hora	Actividad	Lugar
08:00 - 14:00	**Registro de asistentes**	Explanada del Auditorio
08:00 - 09:00	**Ponencia Magistral 1: Aplicaciones de la inteligencia artificial en el paradigma de la educación... ¿evolución o retroceso?** **Dr. David Arnoldo Valtiérrez Angel** Moderador: ----------	Auditorio
09:00 - 10:00	**Ponencia Magistral 2: Aprovechamiento fototérmico solar en acristalamiento aislante integrado con PCM.** **profesor Dong Li** Moderador: ----------	Auditorio
10:00 - 12:00	**Presentación de pósteres**	Explanada del Auditorio

Ponencias Sala 1, Sesión 1: Tecnología 5.0 y Ciencia de Datos Moderador: DCC					
10:00 - 10:20	Virtual	CITCA2025-02-0088	Horizontal Fragmentation of Large-Scale Medical Databases: Scoping Review and Architecture	Jhonatan Barroso García	Sala 1: Auditorio
10:20 - 10:40	Virtual	CITCA2025-02-0149	Application of Industry 5.0 Technology in Water Conservation: Development of a Cross-Platform Educational App Using FlutterFlow and Expo Snack in a Marginalized High School Setting	Verónica Espinoza Romo	
10:40 - 11:00	Presencial	CITCA2025-02-0041	Analysis of SSH Service Logs for the Detection and Prevention of Cyberattacks and Unauthorized Access	David Arnoldo Valtiérrez Angel	
11:00 -11:20	Virtual	CITCA2025-02-0248	Comparative Performance Analysis of Raspberry Pi Pico and ESP32 Microcontrollers under Single- and Multi-Core Processing Conditions.	Ciriaco Ortiz Carrasco	
11:20-11:40	Virtual	CITCA2025-02-0314	ADAS Systems Prototyping: Distributed Architecture with eCAL Implemented on Raspberry Pi	Jaqueline Cárdenas Regalado	

11:40 - 12:00	**RECESO**	
12:00 - 13:00	**Ponencia Magistral 3: Supresión de oscilaciones o vibraciones en sistemas mecánicos diferencialmente planos.** **Dr. Francisco Beltrán Carbajal** **Moderador: ----------**	**Auditorio**
13:00 - 14:00	**Ponencia Magistral 4: ----------** **----------** **Moderador:**	
14:00 - 14:20	**RECESO**	

Ponencias Sala 1, Sesión 2: Salud Moderador: CCI					
14:20 - 14:40	Virtual	CITCA2025-02-0275	Portable Device With Oxygenation And Temperature Sensors For Monitoring Respiratory Diseases	Hannia Marquez Garcia	Sala 1: Auditorio
14:40 - 15:00	Virtual	CITCA2025-02-0278	Mechanical Percussion Device for the Prevention of Myofascial Syndrome in the Back in Patients: A Controlled Trial	Jose Maria Barragan Mendez	
15:00 - 15:40	Presencial	CITCA2025-02-0257	Review of Noninvasive Techniques for the Early Detection of Cognitive Impairment in Older Adults	Martha Betsabé Alvares González	
15:40 - 16:00	Virtual	CITCA2025-02-0279	Wearable Plethysmograph Design	Jorge Eduardo Hernández Banda	
16:00 - 16:20	Virtual	CITCA2025-02-0285	Additive Technology in the Manufacture of Prostheses for Dog Hind Limbs	José Julián Delgado Rodríguez	

Ponencias Sala 2, Sesión 1: Nominados Mejor Artículo de Congreso Moderador: ----					
10:00 - 10:20	Presencial	CITCA2025-02-0035	Battery Pack Sizing and Configuration for an Electric Formula SAE Vehicle	Melchor Alejandro Camacho Guevara	

10:20 - 10:40	Virtual	CITCA2025-02-0308	Comparison of segmentation methods, using SAM2, DeepLabV3+ and a proprietary for orange detection.	Anibal González Tovar	Sala 2: Aula 2102 Edificio del Departamento de Ingeniería Electrónica
10:40 - 11:00	Presencial	CITCA2025-02-0162	Portable Power Prototype Powered by Solar or Wind Energy	Gabriela Yalit Ortiz Lagunas	
11:00 -11:20	Virtual	CITCA2025-02-0317	NDVIcam: Cámara Multiespectral para obtener el índice NDVI y niveles de Nitrógeno en plantas iluminadas con LEDs	Luis Antonio Gama Moreno	
11:20 - 11:40	Presencial	CITCA2025-02-0242	Redesign and Construction of a Multifunctional Reactor for Biofuel Production	Sandro Vazquez Maceda	
11:40 - 12:00	**RECESO**				
12:00 - 13:00	**Ponencia Magistral 3: Supresión de oscilaciones o vibraciones en sistemas mecánicos diferencialmente planos. Dr. Francisco Beltrán Carbajal**				**Auditorio**
13:00 - 14:00	**Ponencia Magistral 4: ---------- ----------**				
14:00 - 14:20	**RECESO**				
Ponencias Sala 2, Sesión 2: Nominados Mejor Artículo de Congreso Moderador: ----					
14:20 - 14:40	Presencial	CITCA2025-02-0183	Enhancing Torque Generation in an In-pipe Savonius Turbine Using an Eye-shaped Flow Deflector: A CFD Study	José Carrasco García	Sala 2: Aula 2102 Edificio del Departamento de Ingeniería Electrónica
14:40 - 15:00	Virtual	CITCA2025-02-0276	Classification Of Emg Signals Using Neural Networks In The Activation Of The Biceps Brachii Muscle	Jesus Misraim Herrera Ramirez	
15:00 - 15:40	Presencial	CITCA2025-02-0338	Exploration of Chaotic Spwm in Reconfigurable Systems: From Logistic MAP to THD Analysis	Hiram Ochoa Sáenz	
15:40 - 16:00	Presencial	CITCA2025-02-0336	Assessing self-regulated learning using cluster analysis applied to MSLQ data in higher education	Miguel Ángel Couoh Novelo	
16:00 - 16:20	Virtual	CITCA2025-02-0277	Prototype of a Wearable Device for Monitoring Oxygenation and Heart Rate in Patients	Guadalupe Azucena Serapio Moro	

Ponencias Sala 3, Sesión 1: Nominados Mejor Artículo de Revista Moderador: ----					
10:00 - 10:20	Presencial	CITCA2025-02-0233	Stress and Strain Analyzer to improve the teaching-learning process of Mechatronics Engineering.	Luis Manuel García Martínez.	Sala 3: Aula 2109 Edificio del Departamento de Ingeniería Electrónica
10:20 - 10:40	Presencial	CITCA2025-02-0178	1D-TiO2 thin films with CdS quantum dots for the production of solar fuels from greenhouse gases (CO2)	Luis Antonio García Contreras	
10:40 - 11:00	Virtual	CITCA2025-02-0223	Implementation of algorithms for data extraction and classification in electrocardiograms	Norberto Urbina Brito	
11:00 -11:20	Presencial	CITCA2025-02-0029	An overview of cohesion metrics in microservices architectures	Samanta LAra Agapito	
11:20 - 11:40	Presencial	CITCA2025-02-0171	Photocatalytic performance of sol–gel synthesized ZnO in methylene blue degradation under direct photocatalysis and photo-Fenton-like conditions	Lorenzo Antonio Cruz Santiago	
11:40 - 12:00	**RECESO**				
12:00 - 13:00	**Ponencia Magistral 3: Supresión de oscilaciones o vibraciones en sistemas mecánicos diferencialmente planos. Dr. Francisco Beltrán Carbajal**				**Auditorio**
13:00 - 14:00	**Ponencia Magistral 4: ---------- ----------**				
14:00 - 14:20	**RECESO**				
Ponencias Sala 3, Sesión 2: Nominados Mejor Artículo de Revista Moderador: ----					
14:20 - 14:40	Presencial	CITCA2025-02-0206	Parameterization and Control Based on a Linear Quadratic Regulator with Complementary Filter of Self-Balancing Robot	Luis Felipe Serna Hernández	Sala 3: Aula 2109 Edificio del Departamento de Ingeniería Electrónica
14:40 - 15:00	Presencial	CITCA2025-02-0018	Measuring Energy, Water Consumption, and Carbon Footprint for each Query Generated by ChatGPT	David Arnoldo Valtiérrez Angel	
15:00 - 15:40	Virtual	CITCA2025-02-0245	Impact of an Aloe vera-Based Coating on the Postharvest Preservation of Raspberries: A Study on Centennial and 1473 Varieties	Juan Pablo Mojica Sánchez	
15:40 - 16:00		CITCA2025-02-0300	Temperature profile in heat sink used in thermoelectric modules		

16:00 - 16:20					

Ponencias Sala 4, Sesión 1: Desarrollo Sostenible Moderador: DCIM					
10:00 - 10:20	Virtual	CITCA2025-02-0079	Trends in new manufacturing methods for sandwich structures and their influence on mechanical properties	Juan Antonio Villa Guerrero	Sala 4: Aula 2111 Edificio del Departamento de Ingeniería Electrónica
10:20 - 10:40	Presencial	CITCA2025-02-0089	A Review on the Impact of Materials on the Mechanical Performance of Energy-Storing-and-Returning (ESR) Foot Prostheses	Sandra Yaraset Vara Salas	
10:40 - 11:00	Virtual	CITCA2025-02-0200	Implementation of New Techonoligies in Aviation and Its Impact on the Environment	Nativitas Juarez Katerine Michelle	
11:00 -11:20	Virtual	CITCA2025-02-0227	Identification of Natural Enemies of Diaphorina Citri in Citrus Fruits	Eder Yosmar Hernandez Hernandez	
11:20 - 11:40	Presencial	CITCA2025-02-0202	Micro-Scale Wind Assessment with Commercial Uavs and IoT Sensors for Comfort and Environmental Management in Morelos.	Héctor Vázquez Chávez	
11:40 - 12:00	**RECESO**				
12:00 - 13:00	**Ponencia Magistral 3: Supresión de oscilaciones o vibraciones en sistemas mecánicos diferencialmente planos. Dr. Francisco Beltrán Carbajal**				**Auditorio**
13:00 - 14:00	**Ponencia Magistral 4: ---------- ----------**				
14:00 - 14:20	**RECESO**				
Ponencias Sala 4, Sesión 2: Inteligencia Artificial Moderador: DCC					
14:20 - 14:40	Presencial	CITCA2025-02-0201	Eye Tracking with CNN and Webcam: A Comparison of Single-Eye Tracking (SET) and Double-Eye Tracking (DET)	Joel Enrique Urquiza Martinez	Sala 4: Aula 2111 Edificio del Departamento de Ingeniería Electrónica
14:40 - 15:00	Virtual	CITCA2025-02-0131	An approach to intelligent packaging selection using computer vision	Víctor Hugo García Espinoza	
15:00 - 15:40	Presencial	CITCA2025-02-0189	Exploring the Use of Generative Artificial Intelligence to Support Requirements Specification in Software Engineering Education.	María Clara Gómez Álvarez	
15:40 - 16:00	Virtual	CITCA2025-02-0080	Intelligent Models for Predicting Phonological Development	Maria Herlinda Castillo Hernández	
16:00 - 16:20	Virtual	CITCA2025-02-0132	Artificial Intelligence Applied To Absorption Heat Pumps	Gerardo Manríquez Pardo	

Ponencias Sala 5, Sesión 1: Gestión del Agua Moderador: CCI					
10:00 - 10:20	Presencial	CITCA2025-02-0198	Mechanical Design of Aquatic Robot Applaed to Collect Solid Waste in Lake Areas of Mexico City	Erick Gael Avila Cruz	Sala 5: Aula 3101 Edificio del Departamento de Ciencias Computacionales
10:20 - 10:40	Virtual	CITCA2025-02-0055	Development of a portable device for in situ monitoring of settleable solids in industrial wastewaters	Manuel de Jesus Cardenas Medina	
10:40 - 11:00	Virtual	CITCA2025-02-0086	Comparación de sales férricas en el proceso foto-Fenton solar para el tratamiento de aguas residuales de la industria textil	Vianney Diaz Blancas	
11:00 -11:20	Virtual	CITCA2025-02-0179	Artificial Wetland as Ecotechnology in the Wastewater Treatment Plant (WWTP) using AutoCAD	Monica Mendoza Kohrs	
11:20 - 11:40		CITCA2025-02-0284	Implications of Slaughterhouse Wastewater: A Socio-environmental Perspective		
11:40 - 12:00	**RECESO**				
12:00 - 13:00	**Ponencia Magistral 3: Supresión de oscilaciones o vibraciones en sistemas mecánicos diferencialmente planos. Dr. Francisco Beltrán Carbajal**				**Auditorio**
13:00 - 14:00	**Ponencia Magistral 4: ---------- ----------**				
14:00 - 14:20	**RECESO**				
Ponencias Sala 5, Sesión 2: Inteligencia Artificial Moderador: ----					
14:20 - 14:40	Virtual	CITCA2025-02-0290	Integrating Advanced Imputation and Genetic Algorithms to Enhance Ovarian Classification	Valeria Maeda Gutiérrez	

14:40 - 15:00	Virtual	CITCA2025-02-0304	Metabolic Biomarkers for Classification of Type 2 Diabetes Using Machine Learning	Pablo Adrian Alonso Rodriguez	Sala 5: Aula 3101 Edificio del Departamento de Ciencias Computacionales
15:00 - 15:40	Presencial	CITCA2025-02-0121	Telematics prototype system for vehicle security and tracking using an embedded System	Agustín Guzmán Cortés	
15:40 - 16:00	Virtual	CITCA2025-02-0203	Comparison of Spectral Representation as Inputs in Multistream Convolutional Neural Networks for Musical Genre Classification	Juan Alfredo Guevara Pérez	
16:00 - 16:20	Virtual	CITCA2025-02-0224	Phonocardiogram Classification with CNN: Beat-Based vs. Fixed-Window Segmentation	Mario Enrique García Luna	

Ponencias Sala 6, Sesión 1: Eficiencia Energética y Energías Renovables Moderador: DIE					
10:00 - 10:20	Presencial	CITCA2025-02-0258	High-efficiency resonant DCX transformer DC-DC converter	Emiliano Vázquez Albarrán	Sala 6: Aula 3102 Edificio del Departamento de Ciencias Computacionales
10:20 - 10:40	Presencial	CITCA2025-02-0292	Comparative Analysis of Solar Unmanned Aerial Vehicles: Fixed-Wing vs. Quadcopter Architectures	Hugo Iván Castrejón Salgado	
10:40 - 11:00	Virtual	CITCA2025-02-0192	Adaptation of an Internal Combustion Engine for the Generation of Electric Power Using Biogas Obtained from Agricultural Biomass	Francisco Javier Godínez García	
11:00 -11:20	Virtual	CITCA2025-02-0234	Identification and control of a binary distillation column for an azeotropic mixture	Moisés Lobatón Montiel	
11:20 - 11:40	Presencial	CITCA2025-02-0302	Statistical Analysis of Precipitation and Electricity Demand in the Eastern Regional Control Management (GCRO)	Alejandro Alvarez Organista	
11:40 -12:00	**RECESO**				
12:00 - 13:00	**Ponencia Magistral 3: Supresión de oscilaciones o vibraciones en sistemas mecánicos diferencialmente planos. Dr. Francisco Beltrán Carbajal**				**Auditorio**
13:00 - 14:00	**Ponencia Magistral 4: ---------- ----------**				
14:00 - 14:20	**RECESO**				
Ponencias Sala 6, Sesión 2: Salud Moderador: CCI					
14:20 - 14:40	Virtual	CITCA2025-02-0104-26-09-2025	Overexpression of human cystatin C in E. coli and and its purification: strategies to remove nucleic acid contamination.	Francisco Reyes Espinosa	Sala 6: Aula 3102 Edificio del Departamento de Ciencias Computacionales
14:40 - 15:00	Presencial	CITCA2025-02-0139-06-10-2025	Literature review of digital cognitive assessment for remote dementia detection	Lluvia Arely López Primo	
15:00 - 15:40	Virtual	CITCA2025-02-0239-12-10-2025	Electronic Radioactive Source Simulator as a Tool for Teaching Nuclear Physics	Adriana Jackeline Hernandez Carbajal	
15:40 - 16:00	Virtual	CITCA2025-02-0323-12-10-2025	Real-time Mexican Sign Language Digital Interpreter for Healthcare	Dulce Casandra González Montalvo	
16:00 - 16:20	Virtual	CITCA2025-02-0341-13-10-2025	Manipulation of Inmoov-Based Prosthesis by Myoelectrical Signals	Guillermo Antonio Vazquez Ortega	